

Rule-Guided Synthetic Data Augmentation

Research proposal (concept note)

May 13, 2026

Abstract

We outline **iSHAP** (interventional SHAP), a data-centric proposal: iteratively synthesize training rows so a fitted predictor better matches lightweight *structural* rules (e.g. monotonicity of an intervention on the outcome) while tracking ordinary hold-out task performance. Feasibility is scored from *interventional* summaries of the model (*do*-style sweeps on held-out covariate rows), not from post-hoc explanation plots alone. Optimization uses a scalar or Pareto-style trade-off between performance and feasibility; augmentation stops when the combined score no longer improves.

1 Motivation

Predictive models often achieve strong metrics yet violate simple domain expectations (e.g. higher insulin should not predict higher future glucose under intervention; higher PM2.5 should not predict higher student grades). Post-hoc explanations do not fix the training distribution. iSHAP proposes to *augment* training data with synthetic points chosen to improve both validation performance and interventional alignment with user-specified rules.

2 Methodology

The pipeline (Figure 1) is a loop with one clear quality criterion: *does adding this synthetic batch make the model both more accurate and more rule-consistent?*

1. **Specify rules.** The user writes a small set of directed expectations $u \rightarrow v$ with a qualitative behavior (e.g. “higher insulin $\Rightarrow$ lower future glucose”, monotone decrease). No numerical effect sizes are required.
2. **Train a baseline.** Fit a predictive model (e.g. LightGBM) on the real training split. Record hold-out performance with the task-appropriate metric (RMSE for regression, accuracy for classification).
3. **Score baseline feasibility.** For each rule $u \rightarrow v$, sweep u over a grid on a fixed set of held-out covariate rows and compute the interventional mean $\bar{f}(g) = \frac{1}{n} \sum_i f(X_{-u}^{(i)}, u=g)$. Compare $\bar{f}$ to the expected qualitative shape; aggregate per-rule alignments into a single feasibility score in $[0, 1]$.
4. **Propose a synthetic batch.** Use a Bayesian optimizer (e.g. Optuna TPE) to sample a batch of candidate rows (X_s, y_s) within feature bounds.

5. **Retrain and re-evaluate.** Fit a fresh model on real data plus the candidate batch; recompute validation loss and feasibility.
6. **Accept or stop.** Combine the two numbers via a weight λ (or Pareto comparison):
 $\text{score} = (1 - \lambda) (1 - \text{normalized loss}) + \lambda \text{feasibility}$.
Append the batch if the combined score strictly improves; otherwise stop.
7. **Finalize and audit.** Train a final model on real data plus all accepted synthetic rows. Predict on test data, re-render the interventional curves, and store curves and JSON summaries for transparent inspection.
8. **Optional feedback.** Rules can be refined from new domain input or from a separate causal-discovery step, restarting the loop with updated expectations.

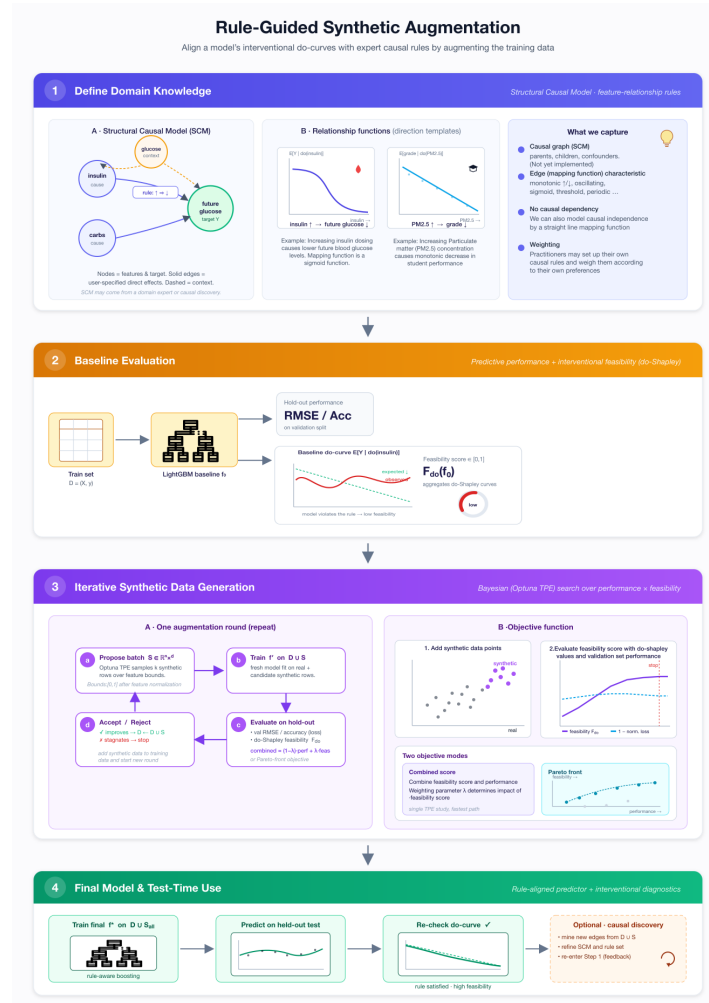


Figure 1: Proposed iSHAP workflow: rules and SCM-style structure → baseline performance and interventional feasibility → synthetic search with accept/stop → final model, test evaluation, and optional discovery.

3 Illustrative examples

Each example below plots the interventional mean prediction $\bar{f}(g) = \frac{1}{n} \sum_i f(X_{-u}^{(i)}, u=g)$ against intervention level g for a single rule, comparing the baseline model (grey, *before* augmentation) to the model trained on real plus iSHAP-generated synthetic rows (red, *after*). A purely monotone curve aligned with the rule sign yields a high per-rule feasibility contribution; jagged or sign-flipping curves do not.

3.1 Example 1: insulin $\rightarrow$ future glucose (d1namo)

Domain expectation. Higher insulin should, on average, push predicted future glucose *down*. The rule is encoded as “monotone decreasing” on the edge insulin $\rightarrow$ future glucose; no numerical slope is fixed.

Reading Figure 2.

- At low insulin ($g \in [0, 4]$) the *baseline* curve is essentially flat with a small bump up to roughly +1.5, locally violating the desired monotone shape.
- The *after* curve in the same region descends gently from about -2 to -4 , restoring monotone decrease at low doses.
- Beyond $g \approx 5$, both curves drop sharply to a plateau around -35 to -37 ; iSHAP slightly deepens the post-step plateau (about 2 units lower) without changing its qualitative shape.

Why this matters. The baseline already captured the high-dose effect but had a non-monotone bump at low doses, the regime where clinical decisions are most sensitive. Augmentation reshapes the low-dose response so it agrees with the rule, while the hold-out metric is reported separately so any accuracy cost is visible.

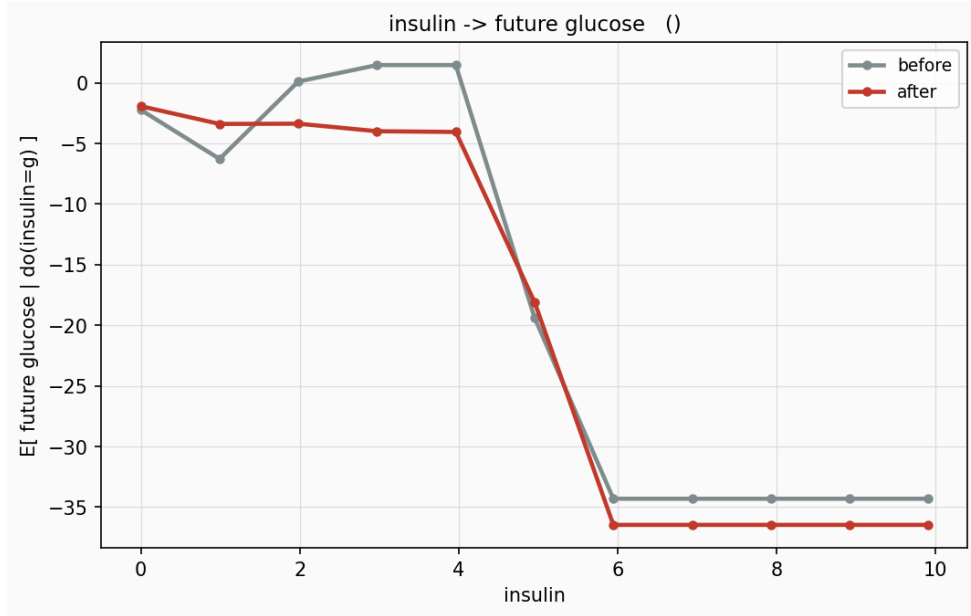


Figure 2: d1namo: interventional mean prediction of future glucose as a function of $\text{do}(\text{insulin}=g)$ on fixed held-out rows. Grey = baseline model, red = after iSHAP augmentation.

3.2 Example 2: PM2.5 $\rightarrow$ grade level (student learning / air quality)

Domain expectation. Higher particulate matter concentration (PM2.5) should, on average, lower predicted academic grade level. The rule is again encoded as monotone decreasing on PM2.5 $\rightarrow$ grade.

Reading Figure 3.

- The baseline curve exhibits a pronounced spike around $g \approx 5.5$ (predicted grade jumps to ≈ 9.46 , well above neighboring levels), and a small dip around $g \approx 8.7$. Two local sign reversals against the rule.
- The after curve removes the spike entirely, replacing it with a flat-to-decreasing plateau near 9.13 for $g \in [4, 6]$ and a smooth gentle decline through $g \in [6, 14]$.
- Both curves agree on a strong drop at the highest PM2.5 level ($g \approx 15$), where the after curve ends slightly higher (≈ 8.70) than the baseline (≈ 8.56) but along a more globally monotone trajectory.

Why this matters. Domain users typically care about the *direction* of response, not its exact magnitude. iSHAP improves global monotonicity (the rule criterion) even when small magnitude differences appear in the tails. The before/after overlay makes such trade-offs explicit rather than hidden behind a single global metric.

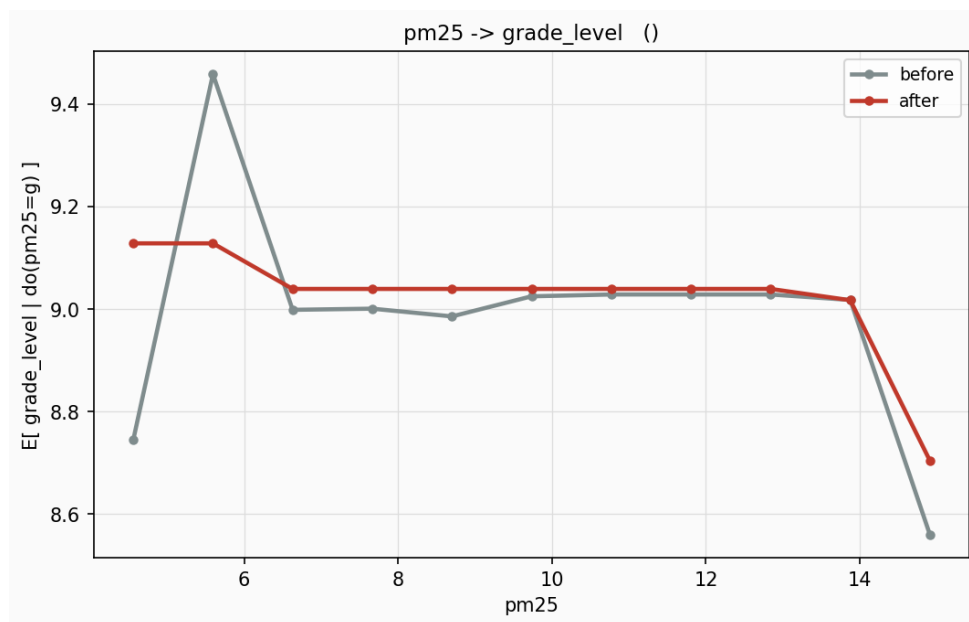


Figure 3: Student learning / air quality: interventional mean prediction of grade level as a function of $\text{do}(\text{PM2.5}=g)$. Grey = baseline, red = after iSHAP augmentation.

4 Key working points

- SCM generation and rule definition
- Better combination of performance and feasibility

- More datasets
- More efficient feasibility scoring
- More efficient synthetic data generation
- Replacing optimization loop with mathematical calculations?